Probability Theory

Starter Problems

1. Suppose that 40 students in Math Club are fans of the Boston Celtics and the other 10 students are fans of the Philadelphia 76ers. If we select two random students, what is the probability there will be 1 Celtics fan and 1 76ers fan?

2. Chloe chooses a real number uniformly at random from the interval $[0, 2017]$. Independently, Laurent chooses a real number uniformly at random from the interval $[0, 4034]$. What is the probability that Laurent's number is greater than Chloe's number? (AMC 12A 2017)

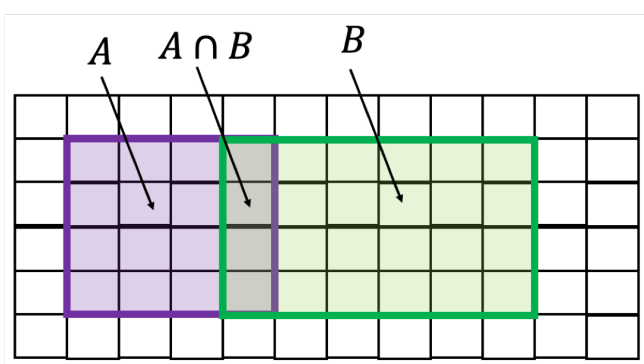
3. A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible? (AMC 10B 2012)

Bayes Theorem

Here, we work with conditional probabilities, which describes the probability of an event given some pre-existing knowledge. In particular, we find the probability $P(A \mid B)$ that an event A occurs given that some event B has already occurred. By **Bayes Theorem**,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{P(B)}$$

The intuition is that we look exclusively among the events where B occurred and see in how many of those events did A also occur. Note that $P(B \mid A)$ represents the probability that B happens given that A has already happened and $P(A \cap B)$ describes the probability of both events. This visual diagram should hopefully provide an additional layer of conceptual understanding.



We also note that for some events A and B, we have the Chain Rule $P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$. This orbits around the multiplicative understanding of probability

that $P(A \cap B)$ is essentially the probability of A times the probability of B given that A happened.

We can extend our grasp of Bayes Theorem via the **Law of Total Probablity**, which stipulates that for events A, B, and A^c (the inverse event where A does not occur), we have the following:

$$P(B) = P(B \cap A) + P(B \cap A^c) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

For example, if A is the event of flipping a heads, A^c is the event of not flipping a heads (tails). This equation makes intuitive sense in that either A or A^c occurs, which means that when B occurs, either B and A ($B \cap A$) jointly occur, or B and A^c ($B \cap A^c$) jointly occur. The rest is Chain Rule.

We can reconfigure Bayes Theorem as

$$\begin{aligned} P(A \mid B) &= \frac{P(B \mid A)P(A)}{P(B)} \\ &= \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A^c)P(A^c)}. \end{aligned}$$

This is basically the probability of both A and B divided by the sum of the probability of both A and B and the probability of both B and not A.

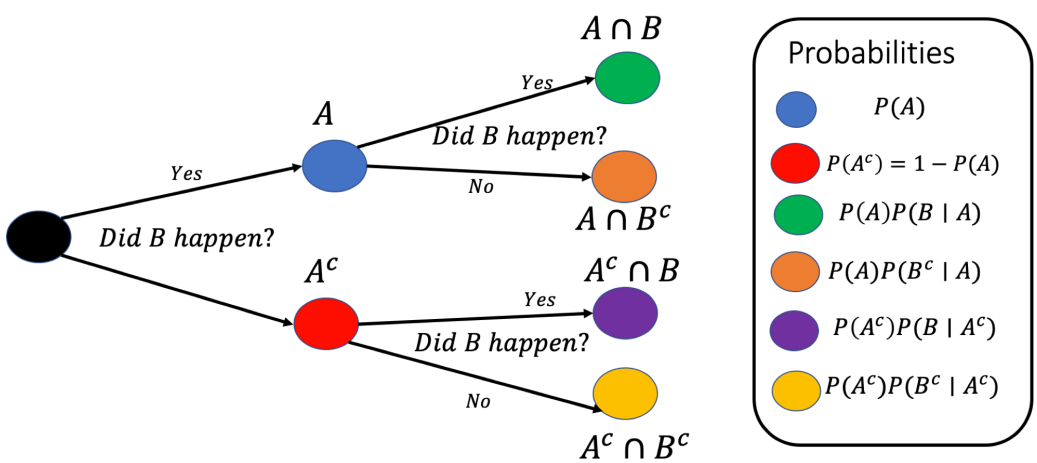


Figure 1: Bayesian Network. We observe that $P(A \mid B)$ is represented by the green probability (A and B both happened) divided by the sum of the green and purple probabilities (event B happened). This is equal to the expression for Bayes Theorem above.

EXAMPLE 1: In a game of *Henry Stickmin*, Henry aims to steal a diamond from a museum and must choose between secret infiltration and large-scale invasion. He is three times as likely to choose infiltration. The win rate for secret infiltration is 60% and the win rate for large-scale invasion is 80%. If Henry succeeds, what is the probability that he won through large-scale invasion?

SOLUTION: As we walk through this solution, note the components of Bayes Theorem that are being implicitly mentioned. He has a 75% chance of selecting infiltration and 25% chance of selecting invasion. Thus, the that Henry succeeds is equal to $0.75 * 0.60 + 0.25 * 0.80 = 0.65$, accounting for both cases—using infiltration $(0.75 * 0.60)$ and using invasion $(0.25 * 0.80)$. The probability of succeeding and using invasion is $0.25 * 0.80$. Thus, the probability that Henry used invasion *given* he succeeded is

$$\frac{P(\text{win } \cap \text{ invasion})}{P(\text{win})} = \frac{0.25 * 0.80}{0.75 * 0.60 + 0.25 * 0.80} = \boxed{\frac{4}{13}}.$$

EXAMPLE 2: Suppose that one of every 500 people in a certain population has a particular disease, which displays no symptoms. A blood test is available for screening for this disease. For a person who has this disease, the test always turns out positive. For a person who does not have the disease, however, there is a 2% false positive rate–in other words, for such people, 98% of the time the test will turn out negative, but 2% of the time the test will turn out positive and will incorrectly indicate that the person has the disease. Let p be the probability that a person who is chosen at random from this population and gets a positive test result actually has the disease. What is p? (AMC 10B 2012)

SOLUTION 1: Let V be the event of a patient having the disease and T be the event of a positive test. We endeavor to find $P(V \mid T)$, the probability that a patient with a positive test result actually has the disease. By Bayes Theorem, we have

$$P(V \mid T) = \frac{P(T \mid V)P(V)}{P(T)}.$$

We have $P(T) = P(T \mid V)P(V) + P(T \mid V^c)P(V^c)$ since disease status is binary (a patient either has it or doesn't have it). We note that a positive test occurs with 100% when a patient has the disease and occurs with 2% chance with a patient does not have the disease (false positive). This means that $P(T \mid V) = 1$ and $P(T \mid V^c) = 0.02$. Additionally, $P(V) = \frac{1}{500}$

since 1 in a 500 people have the virus and $P(V^c) = \frac{499}{500}$. Thus,

$$\begin{aligned} P(V \mid T) &= \frac{P(T \mid V)P(V)}{P(T)} \\ &= \frac{P(T \mid V)P(V)}{P(T \mid V)P(V) + P(T \mid V^c)P(V^c)} \\ &= \frac{1 * \frac{1}{500}}{1 * \frac{1}{500} + 0.02 * \frac{499}{500}} \\ &= \boxed{\frac{50}{549}}. \end{aligned}$$

SOLUTION 2: We can simply consider a sample with 500 people. We expect 1 to have the disease and they will definitely test positive. We expect 499 people to be virus-free and 2% of them to test positive, which equates to $499 * 0.02$ false positive tests. Thus, since $499 * 0.02 + 1$ people will test positive and only 1 of them actually has the disease, the probability of having the disease given a positive test is

$$p = \frac{1}{1 + 499 * 0.02} = \boxed{\frac{50}{549}}.$$

EXAMPLE 3: Two counterfeit coins of equal weight are mixed with 8 identical genuine coins. The weight of each of the counterfeit coins is different from the weight of each of the genuine coins. A pair of coins is selected at random without replacement from the 10 coins. A second pair is selected at random without replacement from the remaining 8 coins. The combined weight of the first pair is equal to the combined weight of the second pair. What is the probability that all 4 selected coins are genuine? (AMC 10A 2011)

SOLUTION: We have 2 imposters scattered among 8 genuine coins—sound familiar? Let A be the event of all 4 coins being genuine and B be the event of the combined weight of the first pair being equal to the combined weight of the second pair. We seek $P(A \mid B)$, which is equal to

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B)}{P(A \cap B) + P(A^c \cap B)}.$$

We consider B first. Among the 4 selected coins, we must either have $\{(g, c), (g, c)\}$, $\{(g, c), (c, g)\}$, $\{(c, g), (g, c)\}$, $\{(c, g), (c, g)\}$, or $\{(g, g), (g, g)\}$ as the sets of selected coin pairs, where g denotes genuine and c counterfeit. We partition these 5 scenarios into their respective categories.

- $A \cap B$: We have $\{(g, g), (g, g)\}$ since all the coins are genuine. The probability of selecting four genuine coins without repalcement is equal to

$$\frac{8}{10} * \frac{7}{9} * \frac{6}{8} * \frac{5}{7} = \frac{1}{3}.$$

- $A^c \cap B$: We omit the case from above, and thus we have $\{(g,c),(g,c)\}$, $\{(g,c),(c,g)\}$, $\{(c,g),(g,c)\}$, or $\{(c,g),(c,g)\}$. Due to symmetry, all of these cases are equally likely. The first one, in particular, occurs with a probability equal to

$$\frac{8}{10} * \frac{2}{9} * \frac{6}{8} * \frac{1}{7} = \frac{1}{45}.$$

  Thus, the total probability is $4 * \frac{1}{45} = \frac{4}{45}$. The individual probabilities are color-coded to display whether they refer to selecting a genuine or counterfeit coin.

Thus, the desired probability $P(A \mid B)$ is

$$P(A \mid B) = \frac{P(A \cap B)}{P(A \cap B) + P(A^c \cap B)} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{4}{45}} = \boxed{\frac{15}{19}}.$$

Geometric Probability

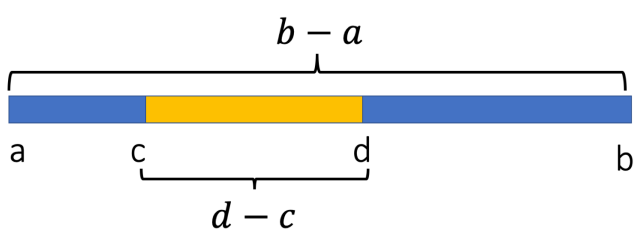
We have primarily focused on **discrete** outcomes so far, meaning that the outcomes belongs to a countable set. For instance, the presence of a disease is a binary variable and thus discrete. The roll of a 6-sided dice and a coin flip also represent discrete variables.

Here, we will laser into **continuous variable**, whose outcomes constitute a spectrum of infinite, uncountable values. An example would be the uniform distribution parametrized by $[a, b]$. This means that our value X is a randomly selected real number in the range $a \leq X \leq b$.

With continuous probabilities, we never describe the probability of a singular number as $P(X = x) = 0$ for any $x \in [a, b]$. More often, we describe the probability that X will lie within some sub-interval. For instance, if we randomly select x in $[a, b]$ and want to find $P(c \leq X \leq d)$ for some $a \leq c < d \leq b$, we have:

$$P(c \leq X \leq d) = \frac{d-c}{b-a}.$$

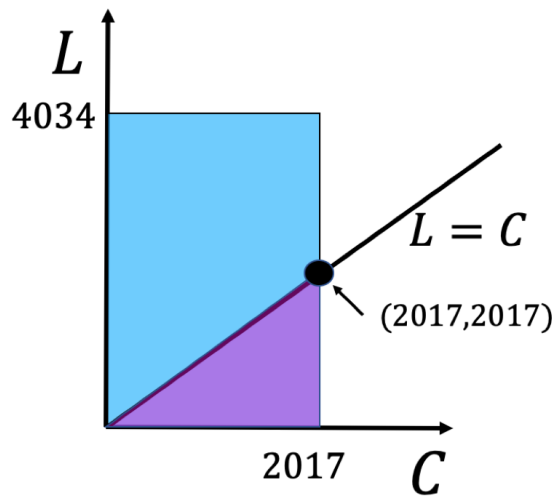
Note that essentially the length of our desired interval is $d - c$ and the length of the total space is $b - a$, so the probability is $\frac{d-c}{b-a}$.



We employ the same intuition for areas and volumes when we want to compute probabilities involving more than one continuous variable. Our general strategy is to geometrically construct the probability space (the possible intervals of the numbers) and find the proportional length/area/volume of the desired space to the total space.

EXAMPLE 4: Chloe chooses a real number uniformly at random from the interval $[0, 2017]$. Independently, Laurent chooses a real number uniformly at random from the interval $[0, 4034]$. What is the probability that Laurent's number is greater than Chloe's number? (AMC 12A 2017)

SOLUTION: Let C be Chloe's number and L be Laurent's number. We have $0 \leq C \leq 2017$ and $0 \leq L \leq 4034$, which forms a rectangle when plotted out. We are interested in the probability that $L > C$, which describes the area above the line $L = C$ as shown below.

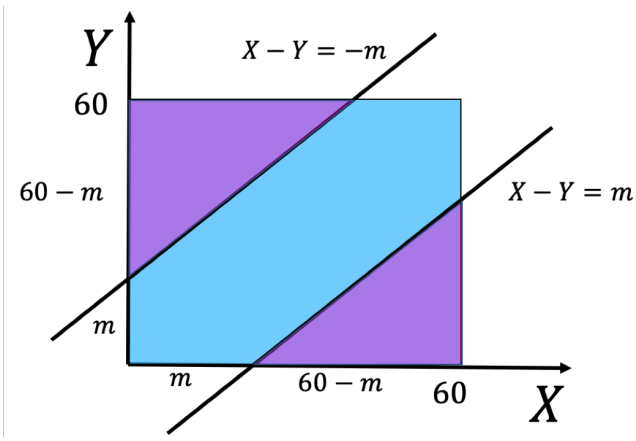


The area of the total space is $4034 * 2017$. The area of the space where Laurent has the higher number is colored in blue, which is simply $4034 * 2017$ minus the purple area, where $L < C$. The purple area is a right triangle that has area $\frac{1}{2}(2017)^2$. Thus,

$$P(L > C) = \frac{[\text{blue}]}{[\text{total}]} = \frac{4034 * 2017 - [\text{purple}]}{4034 * 2017} = \frac{4034 * 2017 - \frac{1}{2}(2017)^2}{4034 * 2017} = \boxed{\frac{3}{4}}.$$

EXAMPLE 5: Two mathematicians take a morning coffee break each day. They arrive at the cafeteria independently, at random times between 9 a.m. and 10 a.m., and stay for exactly m minutes. The probability that either one arrives while the other is in the cafeteria is 40%, and $m = a - b\sqrt{c}$, where a, b, and c are positive integers, and c is not divisible by the square of any prime. Find $a + b + c$. (AIME 1998)

SOLUTION: Let X and Y be the arrival times of the two mathematicians, computed as the number of minutes after 9 AM. The total area of the probability space is $60 * 60$ since $0 \leq X, Y \leq 60$. We are looking for the area that satisfies $|X - Y| \leq m$, which would mean that the two mathematicians arrive within m minutes of each other and thus both meet in the cafeteria. We can graph out the probability space as painted below.



We note that $|X - Y| \leq m$ is equivalent to $-m \leq X - Y \leq m$, which is the blue area between the lines $X - Y = m$ and $X - Y = -m$. The total area is 60^2 and since the purple area is composed of two isosceles right triangles each with base length $60 - m$, we have

$$P(|X - Y| \leq m) = \frac{[\text{blue}]}{[\text{total}]} = \frac{[\text{total}] - [\text{purple}]}{60^2} = \frac{60^2 - (60 - m)^2}{60^2}.$$

This expression reduces to $\frac{120m - m^2}{60^2}$. As stated in the problem, this is equal to 0.4. Thus,

$$\begin{aligned} \frac{120m - m^2}{60^2} &= 0.4 \\ 120m - m^2 &= 1440 \\ m^2 - 120m + 1440 &= 0. \end{aligned}$$

We use the quadratic formula to find $m = \frac{120 \pm \sqrt{120^2 - 4(1)(1440)}}{2} = \frac{120 \pm 24\sqrt{5^2 - 10}}{2} = 60 \pm 12\sqrt{15}$. We see that $60 + 12\sqrt{15}$ is longer than the entire time span between 9 AM and 10 AM, so the correct answer is $60 - 12\sqrt{15}$. Accordingly, we have $a + b + c = 60 + 12 + 15 = \boxed{87}$.

Atlas of Random Variables (RV)

Discrete RV

- *Binomial*: The number of successes X with n independent trials and a probability p of success on each trial. The probability of k successes is

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{1-k}.$$

Binomial random variables are typically used to model experiments such as coin flips: the above equation is equal to probability of k heads given n flips that each have a chance p of heads. If we have a fair coin, then $P(X = k) = \binom{n}{k} 0.5^k (0.5)^{1-k} = \binom{n}{k} (0.5)^n$.

- *Multinomial*: Suppose that we have k possible outcomes with probabilities $p_1,\ p_2, ..., p_k$ from a certain trial. Let x_i denote the number of times that we observe outcome i. If we conduct n trials, the probability of obtaining $x_1,\ x_2, ..., x_k$ counts of each of the k outcomes is given by

$$P(x_1, x_2, ..., x_k) = \binom{n}{x_1, x_2, ..., x_k} * p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k} = \binom{n}{x_1, x_2, ..., x_k} \prod_{i=1}^{n} p_i^{x_i}.$$

  Here, $\binom{n}{x_1, x_2, ..., x_k}$ represents the multinomial coefficient, or $\frac{n!}{x_1! x_2! \cdots x_k!}$. We can use a multinomial distribution to model the arrangements of letters in a word or the frequencies of dice rolls in a not necessarily fair-sided dice. In particular, the number of arrangements of the letters in $LLAMA$, which has 2 Ls, 1 M, and 2 As, is equal to $\frac{5!}{2!1!2!}$.

- *Geometric:* The number of attempts X until we achieve our first success, which consistently occurs with probability p.

$$P(X = k) = (1 - p)^{k-1} p.$$

  For example, the probability that we get our first head with k coin flips is $(1 - p)^{k-1} p$ if $p = P(\text{heads})$.

- *Hypergeometric:* Suppose that we have a bag of N marbles with K red marbles and $N - K$ green marbles. The probability that we draw precisely k red marbles after drawing n marbles without replacement is

$$P(k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

  Note that $\binom{K}{k}\binom{N-K}{n-k}$ describes the number of ways to select k red and $n - k$ green marbles, and $\binom{N}{n}$ describes the number of ways to select n marbles from N total.

- *Negative Binomial:* We model the number of coin flips X until we have obtained r heads with $p = P(\text{heads})$. We have the probability function

$$p(X = k) = \binom{k-1}{r-1} p^r (1 - p)^{k-r}.$$

  We note that this is virtually the Binomial probability of $r - 1$ heads with $k - 1$ coin flips multiplied by the probability p that the kth flip is heads.

- *Poisson*: A Poisson RV describes the number of events X that occur within a certain time or space interval given that an average of λ events occur independently of one another. The probability of $k \geq 0$ events is

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

We can use the Poisson RV to model, for instance, the number of worldwide births in a minute or the number of webpage visits per hour.

Continuous RV

For any continuous distribution, we define the probability density function (PDF) as the relative likelihood of a particular value x being selected. We always have $\int_{-\infty}^{\infty} f(X = x)\, dx = 1$ for any density function f, which essentially means that all probabilities "add" up to 1.

- *Uniform:* For a uniform distribution X in the range $[a, b]$, we have a probability density function of

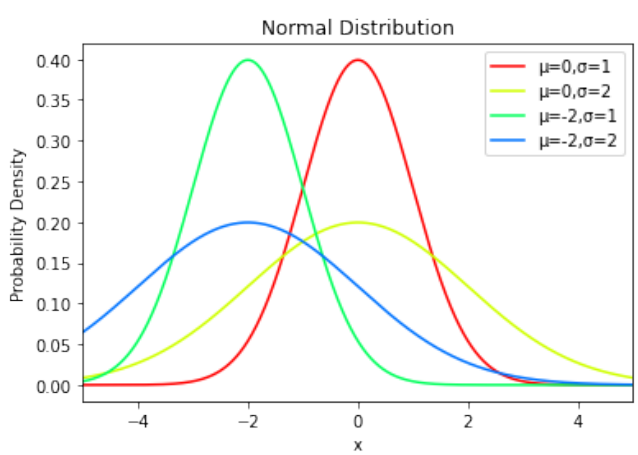
$$f(X = x) = \frac{1}{b - a}.$$

Note that f is not dependent on x, as the probability density is uniform.

- *Normal:* The normal distribution X is widely used in statistics. It can be used to model, for instance, the heights of the human population or temperature fluctuations. If X has a mean of μ and a standard deviation of σ, the probability density function is

$$f(X = x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

We note that as x gets farther away from μ, the probability density f decreases. This is because the highest relative probabilities are near the center μ, as shown below.



Expectation

Let X be a random variable, which is a variable that takes on the outcomes of some probability distribution. We will mostly work with *a priori* distributions whose parameters we already know. For example, if X is the roll of a standard six-sided dice, then $P(X = j) = \frac{1}{6}$ for any $1 \leq j \leq 6$. Since all probabilities are equal and the possible rolls are 1, 2, 3, 4, 5, and 6, we say that the expected or average dice roll is $E[X] = \frac{1+2+3+4+5+6}{6} = 3.5$.

However, we cannot always rely on a uniform distribution. Some outcomes must be weighted higher than others since they have higher probabilities. Formally, if we have a random variable X whose possible outcomes forms the set Ω, we have

$$E[X] = \sum_{x \in \Omega} P(X = x) * x.$$

The expected value is essentially a probability-weighted average of the outcomes. We can see this with more clarity if we revisit the six-sided die.

$$E[X] = \sum_{x \in \{1,2,3,4,5,6\}} \frac{1}{6} * x = \frac{1}{6} * 1 + \cdots + \frac{1}{6} * 6 = 3.5.$$

Suppose that we have two random variables X and Y. It turns out that $E[X+Y] = E[X]+E[Y]$, a principle called the Linearity of Expectation. It is not too complex to prove this but since it involves higher-level mathematics, we will bypass it for now. A proof is offered in the supplement section at the back of the handout.

EXAMPLE 6: Let $p_n(k)$ be the number of permutations of the set $\{1, 2, 3, \ldots, n\}$ which have exactly k fixed points. Prove that $\sum_{k=0}^{n} kp_n(k) = n!$. (International Math Olympiad 1987)

SOLUTION: Recall that a fixed point occurs when i is placed in the ith slot of the permutation. For example, the permutation $\{1, 2, 3\}$ has three fixed points and $\{1, 3, 2\}$ has one fixed point. We can rewrite the desired expression as

$$\sum_{k=0}^{n} k * \frac{p_n(k)}{n!} = 1.$$

Let X be a random variable denoting the number of fixed points in a permutations. We note that $\frac{p_n(k)}{n!}$ describes the probability of $X = k$ fixed points since $n!$ is the total number of permutations and $p_n(k)$ is the number of permutations with k fixed points. Thus, we have

$$\sum_{k=0}^{n} k * p(X = k) = 1.$$

Since X can take on any integer between 0 and n and $0 \leq X \leq n$, then the left hand side is simply $E[X]$. Thus, we need to prove that $E[X] = 1$, where X is the number of fixed points in

a permutation. To finish off the problem, we express X as $X_1 + X_2 + ... + X_n$, where X_i is 1 if slot i has the number i and 0 otherwise. Accordingly,

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_n].$$

It is intuitive to see that $E[X_i] = \frac{1}{n}$ since there are n possible numbers that could be placed at slot i so the probability that a slot receives its own number is equal to $\frac{1}{n}$. Thus,

$$E[X] = \underbrace{\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n}}_{n \text{ times}} = n(\frac{1}{n}) = \boxed{1}.$$

This completes the proof. Note that this is similar to the famous hat problem, where N hats are randomly distributed to N guests and we want to find the expected number of guests $E[X]$ who receive their own hat. Since each person receives their hat with $\frac{1}{N}$ probability, we expect $\frac{1}{N}$ of the guests to get their own hat, which is equal to $\frac{1}{N} * N = 1$ guest.

EXAMPLE 7: At a nursery, 2006 babies sit in a circle. Suddenly, each baby randomly pokes either the baby to its left or to its right. What is the expected value of the number of unpoked babies? (Harvard-MIT Math Tournament 2006)

SOLUTION 1: In order for any baby to be unpoked, the person to their left must have poked left and the person to their right must have poked right. This occurs with $(\frac{1}{2})^2 = \frac{1}{4}$ probability, so we expect $\frac{1}{4}$ of the babies to be unpoked. This equates to $\frac{1}{4} * 2006 = \boxed{501.5}$ unpoked babies.

SOLUTION 2: This is virtually the same solution but slightly more rigorous and formal. Let X be the number of unpoked babies and X_i be a binary variable that is 1 if the ith baby is unpoked and 0 otherwise for $i \in \{1, 2, 3, ..., 2016\}$. Since $X = X_1 + X_2 + \cdots + X_{2006}$, we have

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_{2016}].$$

The probability any baby is unpoked is $(\frac{1}{2})^2 = \frac{1}{4}$ since both neighbors need to poke in the other direction. Thus, $E[X_i] = 1 * p(X_i = 1) + 0 * p(X_i = 0) = p(X_i = 1) = \frac{1}{4}$. Accordingly,

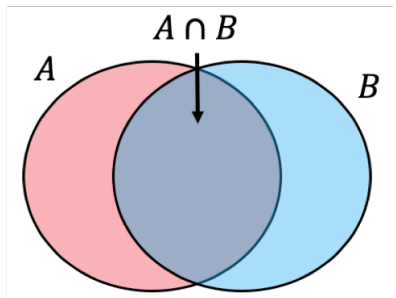
$$E[X] = \underbrace{\frac{1}{4} + \frac{1}{4} + \cdots + \frac{1}{4}}_{2016 \text{ times}} = 2016(\frac{1}{4}) = \boxed{501.5}.$$

Principle of Inclusion-Exclusion

The final destination in our safari of probability theory is the principle of inclusion-exclusion, which is typically represented through Venn diagrams. Essentially, the probability that either event A or B occurs is equal to

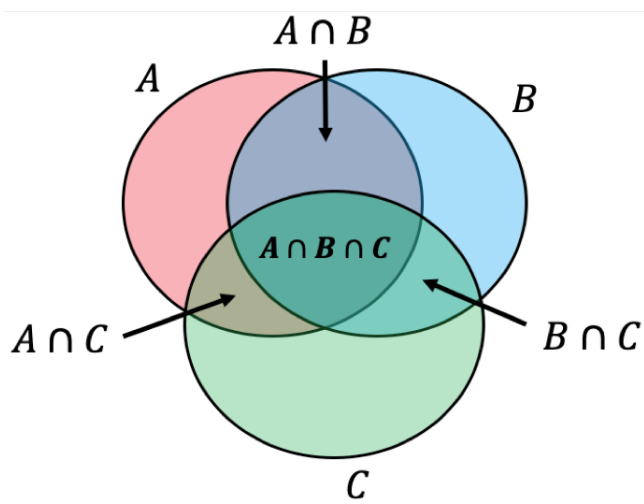
$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Essentially, we sum the probabilities of each event and subtract the probability of both events to avoid overcounting.



We can also use the Principle of Inclusion-Exclusion (PIE) for the size of sets. If A and B are sets with $|A|$ and $|B|$ elements, respectively, and $A \cap B$ has $|A \cap B|$ elements, then the set $A \cup B$ has $|A| + |B| - |A \cap B|$ elements. This works analagously for the other cases as well (with more events). Now, suppose that we now have three events A, B, and C. The probability of either occurring is equal to

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C).$$



This is structurally equivalent to $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$ where $||$ describes the number of elements within a set.

In general, for n events $A_1, A_2, ..., A_n$, the probability of either of the n events occuring is equal to

$$P(\cup_{i=1}^{n} A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) + \cdots + (-1)^{n-1} \sum_{i<\cdots<n} P(\cap_{i=1}^{n} A_i).$$

Notice that we **alternate** the adding and subtracting of probability sums. To find the probability of any of n events occurring, we take the sum of the individual probabilities, subtract the sum of the probabilities of two events at a time, add the sum of the probabilities of three events simultaneously, and alternate repeatedly until we reach the joint probability of all events happening.

Similarly, if we consider n sets $A_1,\ A_2, ..., A_n$, the number of elements within at least one set, or $|\cup_{i=1}^{n} A_i|$, is equal to

$$|\cup_{i=1}^{n} A_i| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| + \cdots + (-1)^{n-1} \sum_{i<\cdots<n} |\cap_{i=1}^{n} A_i|.$$

EXAMPLE 8: In a room, 2/5 of the people are wearing gloves, and 3/4 of the people are wearing hats. What is the minimum number of people in the room wearing both a hat and a glove? (AMC 8 2010)

SOLUTION: Due to the denominators of the fractions, the number of people in the room must be divisible by $\gcd(5, 4) = 20$. Thus, we can write $n = 20k$ where n is the quantity of people in the room and k is some integer. To minimize n, we assume that everyone is wearing either a hat or glove. Thus, there are $\frac{2}{5}(20k) = 8k$ glove wearers and $\frac{3}{4}(20k) = 15k$ hat wearers. If a is the number of people wearing both, by the Principle of Inclusion or Exclusion, we have $8k + 15k - a = 23k - a$ people with either, which is equal to $20k$. Thus, $3k = a$, so we select $k = 1$ and $a = 3$, which leads to a minimum room population of $20k = \boxed{20}$.

EXAMPLE 9: Vernonia High School has 85 seniors, each of whom plays on at least one of the school's three varsity sports teams: football, baseball, and lacrosse. It so happens that 74 are on the football team; 26 are on the baseball team; 17 are on both the football and lacrosse teams; 18 are on both the baseball and football teams; and 13 are on both the baseball and lacrosse teams. Compute the number of seniors playing all three sports, given that twice this number are members of the lacrosse team. (HMMT 2006).

SOLUTION: Let x be the number of seniors playing all three sports, which means that $2x$ seniors play lacrosse. Let F, B, and L be the sets of seniors who play football, baseball, and lacrosse, respectively. The number of seniors playing at least one sport is, by the Principle of Inclusion-Exclusion

$$\begin{aligned} 85 &= |F| + |B| + |L| - |F \cap B| - |B \cap L| - |F \cap L| + |F \cap B \cap L| \\ &= 74 + 26 + 2x - 18 - 13 - 17 + x \\ &= 3x + 52. \end{aligned}$$

Thus, $3x + 52 = 85$, which means that $x = \boxed{11}$, which symbolizes the number of seniors in all three sports.

Problems

1. A fair coin is flipped 10 times. What is the probability of obtaining 5 heads and 5 tails exactly? (*Hint: Binomial Distribution*).

2. Coin A is flipped three times and coin B is flipped four times. What is the probability that the number of heads obtained from flipping the two fair coins is the same? (AMC 10A 2004)

3. Two fair coins are to be tossed once. For each head that results, one fair die is to be rolled. What is the probability that the sum of the die rolls is odd? (Note that if no die is rolled, the sum is 0. (AMC 10B 2008)

4. How many distinct arrangements of the letters in $BOBATEA$ are there?

5. Larry and Julius are playing a game, taking turns throwing a ball at a bottle sitting on a ledge. Larry throws first. The winner is the first person to knock the bottle off the ledge. At each turn the probability that a player knocks the bottle off the ledge is $\frac{1}{2}$, independently of what has happened before. What is the probability that Larry wins the game? (AMC 12B 2015)

6. When flipped, a coin has a probability p of landing heads. When flipped twice, it is twice as likely to land on the same side both times as it is to land on each side once. What is the larger possible value of p? (HMMT 2010)

7. George has two coins, one of which is fair and the other of which always comes up heads. Jacob takes one of them at random and flips it twice. Given that it came up heads both times, what is the probability that it is the coin that always comes up head? (HMMT 2010)

8. A coin is biased in such a way that on each toss the probability of heads is $\frac{2}{3}$ and the probability of tails is $\frac{1}{3}$. The outcomes of the tosses are independent. A player has the choice of playing Game A or Game B. In Game A she tosses the coin three times and wins if all three outcomes are the same. In Game B she tosses the coin four times and wins if both the outcomes of the first and second tosses are the same and the outcomes of the third and fourth tosses are the same. What is the ratio of the probability of winning Game A to the probability of winning Game B? (AMC 12B 2017)

9. Three numbers in the interval $[0, 1]$ are chosen independently and at random. What is the probability that the chosen numbers are the side lengths of a triangle with positive area? (AMC 12A 2016)

10. A circle of radius 1 is randomly placed in a 15-by-36 rectangle $ABCD$ so that the circle lies completely within the rectangle. Given that the probability that the circle will not

touch diagonal AC is m/n, where m and n are relatively prime positive integers. Find $m+n$. (AIME 2004)

11. Dave arrives at an airport which has twelve gates arranged in a straight line with exactly 100 feet between adjacent gates. His departure gate is assigned at random. After waiting at that gate, Dave is told the departure gate has been changed to a different gate, again at random. Let the probability that Dave walks 400 feet or less to the new gate be a fraction $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. (AIME 2010)

12. A dot is marked at each vertex of a triangle ABC. Then, 2, 3, and 7 more dots are marked on the sides AB, BC, and CA, respectively. How many triangles have their vertices at these dots? (HMMT 2006)

13. Amelia has a coin that lands heads with probability $\frac{1}{3}$, and Blaine has a coin that lands on heads with probability $\frac{2}{5}$. Amelia and Blaine alternately toss their coins until someone gets a head; the first one to get a head wins. All coin tosses are independent. Amelia goes first. The probability that Amelia wins is $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $q-p$? (AMC 10A 2017)

14. Three distinct vertices of a cube are chosen at random. What is the probability that the plane determined by these three vertices contains points inside the cube? (AMC 10A 2009)

15. An urn contains 4 green balls and 6 blue balls. A second urn contains 16 green balls and N blue balls. A single ball is drawn at random from each urn. The probability that both balls are of the same color is 0.58. Find N. (AIME I 2014)

16. When 7 fair standard 6-sided dice are thrown, the probability that the sum of the numbers on the top faces is 10 can be written as

$$\frac{n}{6^7},$$

where n is a positive integer. What is n? (AMC 10A 2018)

17. Three distinct integers are selected at random between 1 and 2016, inclusive. Which of the following is a correct statement about the probability p that the product of the three integers is odd? (AMC 10A 2016)

(A) $p < \frac{1}{8}$ **(B)** $p = \frac{1}{8}$ **(C)** $\frac{1}{8} < p < \frac{1}{3}$ **(D)** $p = \frac{1}{3}$ **(E)** $p > \frac{1}{3}$

18. How many 4-digit positive integers (that is, integers between 1000 and 9999, inclusive) having only even digits are divisible by 5? (AMC 12A 2020)

19. A point is chosen at random within the square in the coordinate plane whose vertices are $(0,0), (2020,0), (2020,2020)$, and $(0,2020)$. The probability that the point is within d units of a lattice point is $\frac{1}{2}$. (A point (x,y) is a lattice point if x and y are both integers.) What is d to the nearest tenth? (AMC 12A 2020)

20. Let P be a point chosen uniformly at random in the interior of the unit square with vertices at $(0,0),(1,0),(1,1)$, and $(0,1)$. The probability that the slope of the line determined by P and the point $\left(\frac{5}{8},\frac{3}{8}\right)$ is greater than or equal to $\frac{1}{2}$ can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. (AIME II 2020)

21. A deck of forty cards consists of four 1's, four 2's,..., and four 10's. A matching pair (two cards with the same number) is removed from the deck. Given that these cards are not returned to the deck, let m/n be the probability that two randomly selected cards also form a pair, where m and n are relatively prime positive integers. Find $m+n$. (AIME II 2000)

22. Let S be a square of side length 1. Two points are chosen independently at random on the sides of S. The probability that the straight-line distance between the points is at least $\dfrac{1}{2}$ is $\dfrac{a-b\pi}{c}$, where a, b, and c are positive integers with $\gcd(a,b,c)=1$. What is $a+b+c$? (AMC 10A 2015)

23. Starting with an empty string, we create a string by repeatedly appending one of the letters H, M, T with probabilities $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{1}{4}$, respectively, until the letter M appears twice consecutively. What is the expected value of the length of the resulting string? (HMMT 2016)

24. David is taking a true/false exam with 9 questions. Unfortunately, he doesn't know the answer to any of the questions, but he does know that exactly 5 of the answers are True. In accordance with this, David guesses the answers to all 9 questions, making sure that exactly 5 of his answers are True. What is the probability he answers at least 5 questions correctly? (CMIMC 2020)

25. Newton and Leibniz are playing a game with a coin that comes up heads with probability p. They take turns flipping the coin until one of them wins with Newton going first. Newton wins if he flips a heads and Leibniz wins if he flips a tails. Given that Newton and Leibniz each win the game half of the time, what is the probability p? (HMMT 2010)

26. Justine has a coin which will come up the same as the last flip $\frac{2}{3}$ of the time and the other side $\frac{1}{3}$ of the time. She flips it and it comes up heads. She then flips it 2010 more times. What is the probability that the last flip is heads? (HMMT 2010)

27. Real numbers between 0 and 1, inclusive, are chosen in the following manner. A fair coin is flipped. If it lands heads, then it is flipped again and the chosen number is 0 if the second flip is heads, and 1 if the second flip is tails. On the other hand, if the first coin flip is tails, then the number is chosen uniformly at random from the closed interval $[0,1]$. Two random numbers x and y are chosen independently in this manner. What is the probability that $|x-y|>\frac{1}{2}$? (AMC 10A 2019)

Supplementary Section

Note that this section is a dense forest of combinatorics theory and calculus, so proceed if you have the requisite background and are interested in the material.

Proof of Linearity of Expectation

THEOREM. Let X and Y be two discete random variables. Then $E[X+Y] = E[X] + E[Y]$.

PROOF. We can write the expectation of $X+Y$ as shown below, considering each of the individual cases for $x \in X$ and $y \in Y$ and their corresponding sum $x+y$.

$$\begin{aligned} E[X+Y] &= \sum_{x\in X}\sum_{y\in Y}(x+y)P(X=x\cap Y=y) \\ &= \sum_{x\in X}\sum_{y\in Y}xP(X=x\cap Y=y) + \sum_{x\in X}\sum_{y\in Y}yP(X=x\cap Y=y) \\ &= \sum_{x\in X}x\sum_{y\in Y}P(X=x\cap Y=y) + \sum_{y\in Y}y\sum_{x\in X}P(X=x\cap Y=y) \end{aligned}$$

We note that $\sum_{y\in Y}P(X=x\cap Y=y)$ simply describes the probability of $P(X=x)$ by the Law of Total Probability. Analagously, $\sum_{x\in X}P(X=x\cap Y=y)$ describes the probability of $P(Y=y)$. Thus, we can rewrite the above expression as

$$\begin{aligned} E[X+Y] &= \sum_{x\in X}x\sum_{y\in Y}P(X=x\cap Y=y) + \sum_{y\in Y}y\sum_{x\in X}P(X=x\cap Y=y) \\ &= \sum_{x\in X}xP(X=x) + \sum_{y\in Y}yP(Y=y) \\ &= E[X]+E[Y]. \end{aligned}$$

The final step can be achieved by the very definition of expected value. Note that independence is not required in order for $E[X+Y] = E[X] + E[Y]$ to be true. A similar proof can be achieved for continuous RVs using integrals instead of summations.

Proof of Variance Property

THEOREM. Let X be a discrete random variable. The variance of X is defined as $\mathrm{Var}(X) = E[(X-\mu)^2]$. It is essentially the expected squared deviation from the average μ. Then, $\mathrm{Var}(X)$ is equal to

$$\mathrm{Var}(X) = E[X^2] - E[X]^2$$

PROOF. We start by expanding the right hand side of $\mathrm{Var}(X) = E[(X-\mu)^2]$.

$$\begin{aligned} \mathrm{Var}(X) &= E[(X-\mu)^2] \\ &= \sum_{x\in X}(x-\mu)^2 p(X=x) \\ &= \sum_{x\in X}(x^2-2x\mu+\mu^2)p(X=x). \end{aligned}$$

We partition the above sum as shown below.

$$\begin{aligned}\text{Var}(X) &= \sum_{x\in X}(x^2)p(X=x) + \sum_{x\in X} -2x*\mu*p(X=x) + \sum_{x\in X}(\mu^2)p(X=x) \\ &= \sum_{x\in X}(x^2)p(X=x) - 2\mu\sum_{x\in X} xp(X=x) + (\mu^2)\sum_{x\in X} p(X=x)\end{aligned}$$

We note that $\sum_{x\in X}(x^2)p(X=x)$ is the expected value of random variable X^2. Furthermore, we note that $\sum_{x\in X} xp(X=x)$ is the verbatim definition of $E[X]$, and that $\sum_{x\in X} p(X=x) = 1$ since all the probabilities must add up to 1. Note that $E[X] = \mu$. Thus,

$$\text{Var}(X) = E[X^2] - 2\mu * E[X] + (\mu^2)1 = E[X^2] - 2E[X]^2 + E[X]^2 = E[X^2] - E[X]^2.$$

As a corollary, since $\text{Var}(x) \geq 0$, then $E[X^2] \geq E[X]^2$. As a fun fact, this is a special case of **Jensen's inequality**, which states for some convex function f and random variable X, we have $E[f(X)] \geq f(E[X])$. In this context, we have $f(x) = x^2$.

Probability Density Function

Suppose that continuous random variable X has a probability density function of $f(x)$. The probability that X randomly lies in the interval $[a, b]$ is given by

$$P(a \leq X \leq b) = \int_a^b f(x)\ dx.$$

We note that for any random variable, we have $\int_{-\infty}^{\infty} f(x)\ dx = 1$, which follows the intuition that all probabilities sum to 1. We often work with the **cumulative distribution function** (CDF), which is defined as shown below.

$$P(X \leq x) = F(x) = \int_{-\infty}^{x} f(t)\ dt.$$

The cumulative function essentially outputs the probability that X is less than a certain amount. We can find $P(X > x)$ be simply computing $1 - F(x)$ or $\int_x^{\infty} f(t)\ dt$. Additionally, we note that the density function $f(x)$ is the derivative of the cumulative function $F(x)$ with respect to x, or $\frac{\partial F}{\partial x}$.

For a continuous distribution, we use a slightly distinct formula for $E[X]$, the expectation. In particular, instead of $\sum_x x * p(X = x)$, we have

$$E[X] = \int_{-\infty}^{\infty} xf(x)\ dx.$$

For variance, we are looking for the probability-weighted squared deviation from the mean μ. We can write $\text{Var}(X)$ for continuous X as

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\ dx.$$

Normal Distribution Properties

We aim to find and apply statistical pillars of the normal distribution in real-world settings. The probability density function of a normal distribution X with mean μ and variance σ^2 is

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

We will take the integral of the density function $f(t)$ with bounds $(-\infty, x)$ for some x. This will produce the cumulative distribution function $F(x)$, which is equal to $P(X \leq x)$.

$$\begin{aligned} F(x) &= \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}\ dt \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}}\ dt \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{x} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2}\ dt. \end{aligned}$$

It turns out that the integral $\int_{-\infty}^{x} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2}\ dt$ has no closed-form expression. Statisticians and computer scientists often use pre-tabulated values or powerful computer systems in order to compute cumulative probabilities on the normal distribution. To simplify the above expression, we typically use the Gauss error function, which is defined as

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{0}^{x} e^{-u^2}\ du$$

We can utilize $\operatorname{erf}(x)$ to rewrite and condense the normal CDF equation. We start by performing a substitution on $\operatorname{erf}(x)$ with $u = \frac{t-\mu}{\sqrt{2}\sigma}$. Accordingly,

$$\frac{du}{dt} = \frac{d}{dt}[\frac{t-\mu}{\sqrt{2}\sigma}] = \frac{1}{\sigma\sqrt{2}}.$$

Thus,

$$\operatorname{erf}(\frac{x-\mu}{\sqrt{2}\sigma}) = \frac{2}{\sqrt{\pi}} \int_{\mu}^{x} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2} * \frac{1}{\sigma\sqrt{2}}\ dt = \frac{\sqrt{2}}{\sqrt{\pi}\sigma} \int_{\mu}^{x} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2}\ dt.$$

$$\frac{1}{2}\operatorname{erf}(\frac{x-\mu}{\sqrt{2}\sigma}) = \frac{1}{\sqrt{2\pi}\sigma} \int_{\mu}^{x} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2}\ dt.$$

Finally, we incorporate this into the mainstream equation for $F(x)$ by writing

$$F(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\mu} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2}\ dt + \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mu}^{x} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2}\ dt$$

Since the normal distribution is symmetric about μ—in other words, $f(\mu+t) = f(\mu-t)$ for any t—then the left hand side must be equal to $\frac{1}{2}$. Thus, we have:

$$F(x) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}\sigma} \int_{\mu}^{x} e^{-(\frac{t-\mu}{\sqrt{2}\sigma})^2}\ dt = \boxed{\frac{1}{2} + \frac{1}{2}\operatorname{erf}(\frac{x-\mu}{\sqrt{2}\sigma})}.$$

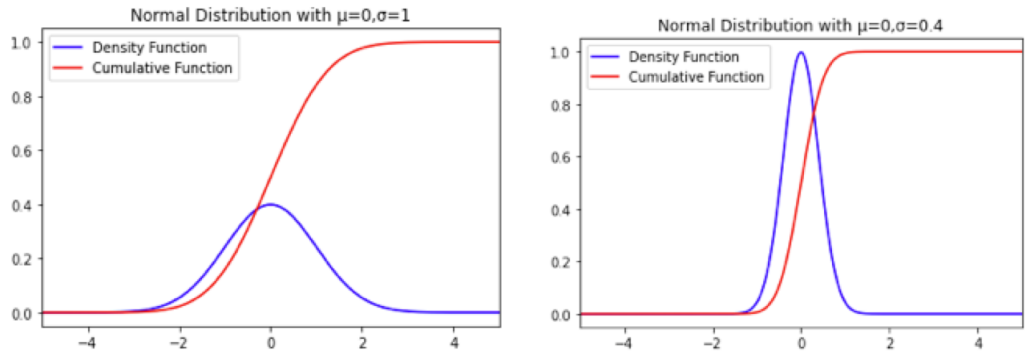


Figure 2: Density and Cumulative Plots for the Normal Distribution

Next, we will show that the normal density function satisfies the notions that $E[X] = \mu$, that the expected value of X is indeed equal to parameter μ. We implement the continuous formula for $E[X]$.

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} (x) f(x) dx \\ &= \int_{-\infty}^{\infty} (x) * \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\ &= = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x e^{-(\frac{x-\mu}{\sqrt{2}\sigma})^2} \ dx. \end{aligned}$$

We perform a u-substitution of $u = \frac{x-\mu}{\sqrt{2}\sigma}$. Thus, we have

$$\frac{du}{dx} = \frac{1}{\sqrt{2}\sigma} \rightarrow dx = \sqrt{2}\sigma \ du.$$

Also, we note that $x = \sqrt{2}\sigma * u + \mu$. By performing substitution, we have

$$\begin{aligned} E[X] &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (\sqrt{2}\sigma * u + \mu) e^{-u^2} * \sqrt{2}\sigma \ du. \\ &= \frac{1}{\sqrt{\pi}} [\sqrt{2}\sigma \int_{-\infty}^{\infty} u e^{-u^2} \ du + \int_{-\infty}^{\infty} \mu e^{-u^2} \ du]. \end{aligned}$$

We note that $\int_{-\infty}^{\infty} e^{-u^2} \ du = \sqrt{\pi}$ due to the properties of Gaussian functions. Additionally, we have $\int_{-\infty}^{\infty} u e^{-u^2} \ du = [-\frac{1}{2} e^{-u^2}]_{-\infty}^{\infty}$, which is essentially 0. More rigorously, we would write

$$\int_{-\infty}^{\infty} u e^{-u^2} \ du = \lim_{n \to \infty} [-\frac{1}{2} e^{-u^2}]_{-n}^{n} = \lim_{n \to \infty} -\frac{1}{2e^{n^2}} + \frac{1}{2e^{n^2}} = 0.$$

By plugging these values into the mainstream equation for $E[X]$, we have

$$E[X] = \frac{1}{\sqrt{\pi}} [0 + \mu\sqrt{\pi}] = \boxed{\mu}.$$

Let n be a number with k distinct prime divisors $p_1, p_2, ..., p_k$. Let A_i be the set of positive integers less than or equal to n and divisible by p_i, which has size $|A_i| = \frac{n}{p_i}$. We consider $|\cup_{i=1}^k A_i|$, which consists of all elements divisible by at least one number in $p_1, p_2, ..., p_k$. Thus,

$$\begin{aligned}
|\cup_{i=1}^k A_i| &= \sum_i |A_i| - \sum_{i,j} |A_i \cap A_j| + \sum_{i,j,k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{k-1} |\cap_{i=1}^k A_i|. \\
&= \sum_i \frac{n}{p_i} - \sum_{i,j} \frac{n}{p_i p_j} + \sum_{i,j,k} \frac{n}{p_i p_j p_k} - \cdots + (-1)^{k-1} \frac{n}{\prod_{i=1}^k p_i} \\
&= n[\sum_i \frac{1}{p_i} - \sum_{i,j} \frac{1}{p_i p_j} + \sum_{i,j,k} \frac{1}{p_i p_j p_k} - \cdots + (-1)^{k-1} \frac{1}{\prod_{i=1}^k p_i}].
\end{aligned}$$

We aim to find $\Phi(x)$, which represents how many numbers are relatively prime with n. This is equal to $n - |\cup_{i=1}^k A_i|$, as we subtract any number $j \leq n$ that shares a prime factor with n.

$$\begin{aligned}
\Phi(x) &= n - |\cup_{i=1}^k A_i| \\
&= n[1 - \sum_i \frac{1}{p_i} + \sum_{i,j} \frac{1}{p_i p_j} - \sum_{i,j,k} \frac{1}{p_i p_j p_k} + \cdots - (-1)^{k-1} \frac{1}{\prod_{i=1}^k p_i}] \\
&= n(1 - \frac{1}{p_1}) \cdots (1 - \frac{1}{p_k}) = n \prod_{i=1}^k (1 - \frac{1}{p_k}).
\end{aligned}$$